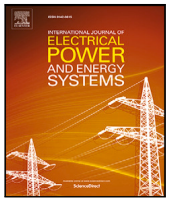
Electrical Power and Energy Systems 156 (2024) 109751



Contents lists available at ScienceDirect

International Journal of Electrical Power and Energy Systems

journal homepage: www.elsevier.com/locate/ijepes



Efficiency evaluation and productivity analysis of complex electric power systems in China: A directional slacks-based network data envelopment analysis approach

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ARTICLE INFO

Keywords:
Data envelopment analysis
Network
Directional slacks-based measure
Luenberger index
Complex electric power system

ABSTRACT

The paper introduces a robust methodology for evaluating complex electricity system efficiency, making several contributions. It systematically decomposes the electric power system into various processes, including power generation, transmission, distribution, and power consumption, with parallel sub-processes for production and domestic power consumption. Employing data envelopment analysis (DEA), the paper develops a network directional slacks-based (DSB) model to evaluate the whole system and the internal processes. The model is theoretically proven to be feasible, providing efficiency scores and identifying inefficiencies in the system and its processes. A noteworthy addition is the introduction of the DSB green Luenberger productivity index and its decomposition, offering an analysis of productivity changes in electricity systems. Empirical analysis covering 30 Chinese provinces from 2013 to 2019 uncovers the performance variations among provinces and processes, pointing out inefficiencies and changes in technology and efficiency over time. The empirical findings aim to support efforts in enhancing the overall efficiency of China's interprovincial power system.

1. Introduction

Electric power is a kind of clean secondary energy, which is widely used because of its convenience and high efficiency. It has great influence on national economic development and people's living standard. Recently, the proportion of electricity in China's terminal energy consumption has greatly increased. In order to reduce carbon dioxide (CO_2) emissions and other pollutant emissions, China actively develops clean energy. However, China still faces the situation that the energy consumption of the electric power industry is dominated by coal. Reducing energy consumption and pollutant emissions in the electric power industry will inevitably affect its overall performance. Therefore, it is necessary to establish a comprehensive performance evaluation method for the electric power industry from the perspectives of energy, environment and economy, etc., so as to provide support for sustainable development.

In order to reduce environmental pollution and energy consumption, numerous studies on the energy and environmental efficiency of the electric power industry are carried out. Among them, data envelopment analysis (DEA) [1] is a very popular efficiency assessment method as it processes undesirable polluting outputs of decision making units (DMUs) well. In view of this advantage, DEA is widely used in the evaluation of energy and environmental efficiency in power. Shu et al. [2] utilized the CCR model [1] in DEA to assess the efficiency of electric power consumption in 29 Chinese provinces from 2001 to 2007. They divided these provinces into four regions, and then analyzed the influencing factors of regional differences. Sueyoshi and Goto [3] measured the environmental efficiency of fossil fuel power plants by using a DEA radial measurement under natural and managerial disposability. Sueyoshi and Goto [4] evaluated the environmental efficiency of American coal-fired power plants and analyzed the advantages and disadvantages of radial models and non-radial models in DEA environmental assessment. Sueyoshi et al. [5] analyzed the environmental efficiency of coal-fired power plants in the United States from 1995 to 2007, and confirmed that Clean Air Act and environmental protection policies had good effects on the environmental efficiency of power plants. Sueyoshi and Goto [6] used DEA to evaluate undesirable congestion under natural disposability and desirable congestion under managerial disposability in the US power industry. Chen et al. [7] considered the game relationship between DMUs and the influence of environmental constraints on the power efficiency, and adopted the

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https://doi.org/10.1016/j.ijepes.2023.109751
Received 26 September 2023; Received in revised form 22 November 2023; Accepted 21 December 2023
Available online 27 December 2023
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DEA cross-efficiency evaluation approach based on the game relationship to study the changes of the power energy efficiency in China. Zurano-Cervelló [8] proposed a novel DEA method to compare and optimize electricity mix based on sustainability criteria and applied it to 28 EU member states. Chen et al. [9] adopted DEA to establish an energy congestion model in order to detect the technical failure and CO_2 emission reduction effect of the coal-fired power generation industry in China.

Note that the above studies only analyze the efficiency of some processes in electricity systems, or all of them use the traditional *black-box* DEA model which ignores the relationship between internal production processes and their influence on the system made up of them. So the efficiency of the whole electricity system is not measured accurately. Network DEA [10] can solve this issue so that it is widely used to evaluate the efficiency of the electric power industry. Xie et al. [11] divided 30 provinces in China into three categories based on the thermal power ratio. Then, a two-stage network DEA model was proposed to evaluate their environmental efficiency. Mahmoudabadi et al. [12] proposed a network the slack-based measure (SBM) [13] model and used it to evaluate 14 electricity companies in Iran. Considering sustainability criteria in economic, social and environmental aspects, Khanjarpanah et al. [14] proposed a new algorithm based on double-boundary network DEA model to determine suitable locations for hybrid power plants. Petridis et al. [15] took profit maximization and cost minimization as the goal of distribution companies and adopted the network DDF model to evaluate the profit efficiency of Turkish distribution network companies. Alizadeh et al. [16] proposed a new dynamic network DEA model to evaluate the performance of complex electricity generation systems by dividing the generic structure of the system to several components, including supplier, plant owner, market manager, consumer, distribution company, transmission, foreign consumer and foreign supplier. Tavassoli et al. [17] proposed a network DEA model for assessing the sustainability of Iran's complex electricity distribution system by dividing it to three stages, involving generation, transmission, and distribution. Zhang et al. [18] designed a two-stage DEA prospect cross-efficiency models to analyze China's electricity generation-transmission system.

While numerous scholars utilize network DEA for evaluating electric power system efficiency, there are no studies that divide the entire complex power system into the processes of power generation, transmission and distribution, production and domestic power to systematically evaluate the efficiency, which can take into account energy, economy and environment comprehensively. Therefore, it is necessary to construct such a network DEA model to completely evaluate the efficiency of electricity generation to consumption in production and life, and identify the detailed inefficient processes as well as factors, so as to provide a theoretical basis for improving power efficiency and reducing carbon dioxide emissions and energy consumption. To achieve this, we do the following researches:

First, we start by breaking down the electric system into sequential processes: power generation, transmission and distribution, and power consumption, aligning with the current state of the electric power system. Recognizing a significant difference between production and domestic power consumption, we further divide the power consumption process into two parallel sub-processes: production power consumption and domestic power consumption, enabling a more detailed analysis. What distinguishes our study is the consideration of this complex network structure within the power system. Existing research has not thoroughly explored this aspect. Particularly noteworthy is the absence of studies that examine the differences between production and domestic electricity consumption separately. This represents a novel approach to evaluating the efficiency of the power system in a comprehensive manner.

Second, utilizing this decomposition, we introduce a network directional slacks-based (DSB) DEA model, along with corresponding efficiency scores, to assess the overall efficiency of electric power systems and their internal processes. This novel model enables the identification of specific inefficient processes and inefficient input–output variables within the power system. In comparison to conventional network DEA models, our model possesses several notable advantages: it maintains linearity without necessitating any transformations; it effectively captures all excesses of inputs and undesirable outputs, along with output shortages; and the weights assigned to each input–output variable can be predetermined based on the practical needs of decision-makers, thereby enhancing its general applicability.

Third, building upon the green Luenberger productivity index (LPI) introduced by [19] and our proposed network DSB model, we present a DSB green Luenberger productivity index (DPI). The DPI for the electric power system is decomposed into a weighted sum of efficiency change indexes and technology change indexes for internal processes. This decomposition allows for a detailed analysis of the reasons behind productivity changes in electric power systems. In comparison to the Malmquist productivity index (MPI) and traditional LPI, our DPI relies on the DSB model. This allows DMUs to assess efficiency in a specific direction, taking into account the inefficiency of slacks, and accommodating the actual preferences of each index. As a result, our approach has broader applicability.

Finally, the proposed model and indexes are then applied to assess the performance of electric power systems across 30 Chinese provinces from 2013 to 2019, inclusive of an analysis of productivity changes in both the systems and individual processes.

The rest of the paper is structured as follows. Section 2 gives the decomposition of electricity system and input–output variables. Based on this, our network DSB model for evaluating the electric power system is constructed. In addition, the efficiency scores and the DPI indexes as well as their decompositions are also provided. In Section 3, we apply our network DEA model and indexes to evaluate the efficiency of electric power systems and internal processes in 30 China's provinces, and analyze the productivity changes based on the resulting data. Section 4 offers conclusions and future research.

2. Method

Various DEA models exist for evaluating specific aspects of electric power systems such as production, consumption, and transmission and distribution. However, there has been a notable absence of studies analyzing the efficiency from the perspective of the entire electric power system. In this section, we address this gap by decomposing the structure of the power system and introducing a network DEA model to evaluate its overall efficiency. Our work focuses on the decomposition of the power system, representing a novel contribution to the field. The proposed network DEA model enables a more comprehensive efficiency assessment in both the entire power systems and their internal processes. To better study the efficiency changes of the whole power systems and internal processes, we also propose the productivity index DPI.

2.1. Decomposition of electric power system and indexes

According to the actual flow of electricity generation, transmission, distribution and consumption, we treat the complex electric system as a series system consisting of three processes: power generation, transmission and distribution, and power consumption. Since the way of power consumption in production and life is very different, we regard the power consumption process as a parallel system composed of two sub-processes[1]: production power consumption and domestic power consumption. See Fig. 1 for the detailed structure of the complex electric system of each province.

[1] For the sake of expression, we also call sub-processes *processes* later in this paper.

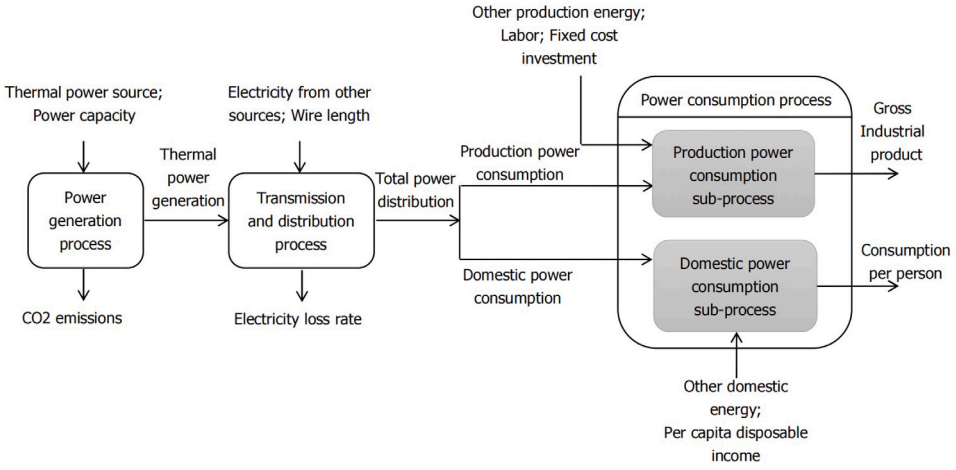


Fig. 1. The detailed decomposition of the electric power system.

To distinguish variables in different processes, we use 1, 2, 3, 31 and 32 on the right superscript to represent the power generation process, transmission and distribution process, power consumption process, production power consumption process and domestic power consumption process, respectively. The first process of the electric power system is the power generation process. Most of the electricity in China is generated by thermal power generation, and other types of power generation are quite different from thermal power generation, especially wind power, solar power and hydropower, which are highly related to the geographical environment of the region. Therefore, the power generation process considered in this paper is the thermal power generation one. According to the electricity-related DEA studies [3–6,9], we select the thermal power source (X_1^1) and power capacity (X_2^1) of thermal power generation as the inputs of the power generation process, and the thermal power generation ($Z^{(1,2)}$) as the intermediate product, which is treated as a desirable output in the first system and meanwhile as an input in the second process. The pollutants produced in the process of thermal power generation include waste water, nitrides, sulfides and CO_2. Because we lack the data on waste water, nitrides and sulfides emitted from thermal power generation in many provinces, and CO_2 emissions can be calculated from the energy used in thermal power generation, CO_2 emissions (B_1^1) are used as the unique undesirable output of thermal power generation process.

The electric energy generated by thermal power generation and other ways needs to be transmitted, transformed and distributed before it can be used. Because transmission, transformation and distribution are closely related, we unify them into one process called transmission and distribution process in this paper, and regard it as the second process of the complex electric power system. Based on the studies on transmission and distribution efficiency [15,16], we select the wire length (X_1^2) and the transmitted electric energy as the inputs, the electricity loss rate (B_1^2) as the undesirable output, and the total power consumption ($Z^{(2,3)}$) as the unique output, which actually is an intermediate product since it is also an input of the power consumption process. Note that the transmitted electric energy includes the thermal power generation from the first process ($Z^{(1,2)}$) and the electricity from other sources (X_2^2), which include electricity purchased from other provinces and generated by other electric power generation ways. The thermal power generation of some provinces is enough to meet the electric energy demand of the whole province. For these provinces, the index of the electricity from other sources (X_2^2) is given a zero value, and the corresponding inefficiency measurement is also given a zero value. The total power consumption ($Z^{(2,3)}$) is all the electric power consumed by the electric power consumption terminal. That is, the sum of production power consumption ($Z^{(2,31)}$) and domestic power consumption ($Z^{(2,32)}$). So, $Z^{(2,3)} = Z^{(2,31)} + Z^{(2,32)}$ establishes.

The final process in the complex electric power system is the power consumption. Considering the great difference between production power consumption and domestic power consumption, we treat the power consumption process as a parallel system composed of the production power consumption process and the domestic power consumption process, so as to make a more detailed analysis. Since the existing studies do not consider the processes of production power consumption and domestic power consumption separately, there is no direct literature reference for the corresponding indicators of these two processes.

The production power consumption process refers to the one in which people use electric energy in various industrial activities to get benefits. In this process, we refer to the literature on DEA-based evaluations of economic, environmental, and energy efficiency [20–23], to guide our selection of input–output indexes. The inputs include other production energy (X_1^{31}), labor (X_2^{31}), fixed cost investment (X_3^{31}) and the production power consumption ($Z^{(2,31)}$), which is distributed from the transmission and distribution process to production activities. There is only one output, which is the gross industrial product (Y_1^{31}).

The domestic power consumption process refers to the one that people improve the quality of their daily life by using electricity. Considering that people use the domestic power consumption ($Z^{(2,32)}$), which is distributed from the transmission and distribution process to people's daily life, other domestic energy (X_1^{32}), and the money they pay for their daily lives to improve and pursue a better life. We select per capita disposable income (X_2^{32}) to represent the amount of money people can pay for life. Because consumption per person (Y_1^{32}) reflects people's living standards, we choose it as the output of this process.

2.2. Network DEA model of evaluating the electric power system

The variables used in what follows are summarized as below:

n: the number of DMUs.

k: the evaluated DMU k ($k \in \{1, \ldots, n\}$).

I^p, R^p, C^p: the number of inputs, outputs and undesirable outputs in process p ($p \in \{1, 2, 3, 31, 32\}$), respectively. According to Fig. 1, we have $I^1 = 2$, $C^1 = 1$, $I^2 = 2$, $C^2 = 1$, $I^{31} = 3$, $R^{31} = 1$, $I^{32} = 2$, $R^{32} = 1$.

x_{ij}^p: the ith ($i = 1, \ldots, I^p$) input of DMU j in process p ($p \in \{1, 2, 3, 31, 32\}$).

y_{rj}^p: the rth ($r = 1, \ldots, R^p$) output of DMU j in process p ($p \in \{1, 2, 3, 31, 32\}$).

b_{cj}^{p}: the cth ($c = 1, \ldots, C^{p}$) undesirable output of DMU j in process p ($p \in \{1, 2, 3, 31, 32\}$).

$z_{j}^{(p,q)}$: the intermediate product[2] of DMU j, produced by process p and utilized by process q ($p, q \in \{1, 2, 3, 31, 32\}$).

$\alpha_{i}^{p}, \beta_{r}^{p}, \sigma_{c}^{p}, \gamma_{x}^{p}, \gamma_{y}^{p}$: the inefficiency measures of inputs, outputs, undesirable outputs, intermediate products as inputs and intermediate products as outputs in processes p ($p \in \{1, 2, 3, 31, 32\}$), respectively.

ω^{p}: the inefficiency weight of process p ($p \in \{1, 2, 3\}$) in the system efficiency, satisfying $\sum_{p=1}^{3} \omega^{p} = 1$.

ρ^{p}: the inefficiency weight of process p ($p \in \{31, 32\}$) in the efficiency of the power consumption process, satisfying $\rho^{31} + \rho^{32} = 1$.

$g_{ik}^{xp}, g_{rk}^{yp}, g_{ck}^{bp}, g_{k}^{(p,q)}$: the direction variables corresponding to inputs, outputs, undesirable outputs in process p and those of the intermediate product from process p to process q ($p, q \in \{1, 2, 3, 31, 32\}$).

$\mu_{i}^{\alpha p}, \mu_{r}^{\beta p}, \mu_{c}^{\sigma p}, \mu^{\gamma_x p}, \mu^{\gamma_y p}$: the weights of inefficiency measures with respect to inputs, outputs, undesirable outputs, intermediate products as inputs and intermediate products as outputs in process p ($p \in \{1, 2, 3, 31, 32\}$), respectively, which are predetermined by decision makers and satisfy

$$\begin{aligned}&\sum_{i=1}^{I^1} \mu_i^{\alpha 1} + \mu_y^{\gamma 1} + \sum_{c=1}^{C^1} \mu_c^{\sigma 1} = \sum_{i=1}^{I^2} \mu_i^{\alpha 2} + \mu_x^{\gamma 2} + \mu_y^{\gamma 2} + \sum_{c=1}^{C^2} \mu_c^{\sigma 2} \\ &= \sum_{i=1}^{I^{31}} \mu_i^{\alpha 31} + \mu^{\gamma_x 31} + \sum_{r=1}^{R^{31}} \mu_r^{\beta 31} \\ &= \sum_{i=1}^{I^{32}} \mu_i^{\alpha 32} + \mu^{\gamma_x 32} + \sum_{r=1}^{R^{32}} \mu_r^{\beta 32} = 1. \end{aligned} \tag{1}$$

There is no actual preference for any weight in this paper so that according to (1), we preset them in this paper as below:

$$\begin{aligned}&\mu_i^{\alpha 1} = \mu_y^{\gamma 1} = \mu_c^{\sigma 1} = \frac{1}{I^1 + C^1 + 1} = \frac{1}{4}, \mu_i^{\alpha 2} = \mu_x^{\gamma 2} = \mu_y^{\gamma 2} = \mu_c^{\sigma 2} = \frac{1}{I^2 + C^2 + 1 + 1} = \frac{1}{5}, \\ &\mu_i^{\alpha 31} + \mu^{\gamma_x 31} + \mu_r^{\beta 31} = \frac{1}{I^{31} + R^{31} + 1} = \frac{1}{5}, \mu_i^{\alpha 32} + \mu^{\gamma_x 32} + \mu_r^{\beta 32} = \frac{1}{I^{32} + R^{32} + 1} = \frac{1}{4}, \\ &\omega^1 = \omega^2 = \omega^3 = \frac{1}{3}, \rho^1 = \rho^2 = \frac{1}{2}. \end{aligned} \tag{2}$$

The directional slacks-based (DSB) measure [24] is a non-radial DEA model that can better express the efficiency of weak efficient DMU than radial DEA models. Compared with the SBM model [13], the DSB model is linear without any transformation and can be solved more easily. This advantage contributes to a more straightforward resolution of the network model constructed later. With the introduction of the directional vector concept, DSB offers greater flexibility, enabling the assessment of technical efficiency in various directions. This aids managers in providing more targeted improvement measures. Furthermore, another benefit of DSB is its adaptability in the presence of negative data, as it allows for the selection of non-negative directional vectors, making it more universal compared to the SBM model. Based on DSB, we construct the following network DSB model to assess the overall inefficiency of the complex electric power system shown in Fig. 1.

$$\begin{aligned} D_k^* = \max\ & \omega^1 \Big(\sum_{i=1}^{I^1} \mu_i^{\alpha 1} \alpha_i^1 + \mu^{\gamma_y 1} \gamma_y^1 + \sum_{c=1}^{C_1} \mu_c^{\sigma 1} \sigma_c^1 \Big) \\ &+ \omega^2 \Big(\sum_{i \in \{i | g_{ik}^{x2} \neq 0\}} \mu_i^{\alpha 2} \alpha_i^2 + \mu^{\gamma_x 2} \gamma_x^2 + \mu^{\gamma_y 2} \gamma_y^2 \\ &+ \sum_{c=1}^{C^2} \mu_c^{\sigma 2} \sigma_c^2 \Big) + \omega^3 \Big[\rho^{31} \Big(\sum_{i=1}^{I^{31}} \mu_i^{\alpha 31} \alpha_i^{31} + \mu^{\gamma_y 31} \gamma_y^{31} + \sum_{r=1}^{R^{31}} \mu_r^{\beta 31} \beta_r^{31} \Big) \\ &+ \rho^{32} \Big(\sum_{i=1}^{I^{32}} \mu_i^{\alpha 32} \alpha_i^{32} + \mu^{\gamma_x 32} \gamma_x^{32} + \sum_{r=1}^{R^{32}} \mu_r^{\beta 32} \beta_r^{32} \Big) \Big] \end{aligned} \tag{3a}$$

$$\text{s.t.} \sum_{j=1}^{n} \lambda_j^1 x_{ij}^1 \leq x_{ik}^1 - \alpha_i^1 g_{ik}^{x1}, \ i = 1, \ldots, I^1, \tag{3b}$$

$$\sum_{j=1}^{n} \lambda_j^1 z_j^{(1,2)} \geq z_k^{(1,2)} + \gamma_y^1 g_k^{(1,2)}, \tag{3c}$$

$$\sum_{j=1}^{n} \lambda_j^1 b_{cj}^1 \leq b_{ck}^1 - \sigma_c^1 g_{ck}^{b1}, \ c = 1, \ldots, C^1, \tag{3d}$$

$$\sum_{j=1}^{n} \lambda_j^2 x_{ij}^2 \leq x_{ik}^2 - \alpha_i^2 g_{ik}^{x2}, \ i = 1, \ldots, I^2, \tag{3e}$$

$$\sum_{j=1}^{n} \lambda_j^2 z_j^{(1,2)} \leq z_k^{(1,2)} - \gamma_x^2 g_k^{(1,2)}, \tag{3f}$$

$$\sum_{j=1}^{n} \lambda_j^2 z_j^{(2,3)} \geq z_k^{(2,3)} + \gamma_y^2 g_k^{(2,3)}, \tag{3g}$$

$$\sum_{j=1}^{n} \lambda_j^2 b_{cj}^2 \leq b_{ck}^2 - \sigma_c^2 g_{ck}^{b2}, \ c = 1, \ldots, C^2, \tag{3h}$$

$$\sum_{j=1}^{n} \lambda_j^{31} x_{ij}^{31} \leq x_{ik}^{31} - \alpha_i^{31} g_{ik}^{x31}, \ i = 1, \ldots, I^{31}, \tag{3i}$$

$$\sum_{j=1}^{n} \lambda_j^{31} z_j^{(2,31)} \leq z_k^{(2,31)} - \gamma_y^{31} g_k^{(2,31)}, \tag{3j}$$

$$\sum_{j=1}^{n} \lambda_j^{31} y_{rj}^{31} \geq y_{rk}^{31} + \beta_r^{31} g_{rk}^{y31}, \ r = 1, \ldots, R^{31}, \tag{3k}$$

$$\sum_{j=1}^{n} \lambda_j^{32} x_{ij}^{32} \leq x_{ik}^{32} - \alpha_i^{32} g_{ik}^{x32}, \ i = 1, \ldots, I^{32}, \tag{3l}$$

$$\sum_{j=1}^{n} \lambda_j^{32} z_j^{(2,32)} \leq z_k^{(2,32)} - \gamma_x^{32} g_k^{(2,32)}, \tag{3m}$$

$$\sum_{j=1}^{n} \lambda_j^{32} y_{rj}^{32} \geq y_{rk}^{32} + \beta_r^{32} g_{rk}^{y32}, \ r = 1, \ldots, R^{32}, \tag{3n}$$

$$\sum_{j=1}^{n} \lambda_j^{p} = 1, \lambda_j^1, \lambda_j^2, \lambda_j^{31}, \lambda_j^{32} \geq 0, \ \forall j, p \tag{3o}$$

$$\sum_{j=1}^{n} \lambda_j^1 z_j^{(1,2)} = \sum_{j=1}^{n} \lambda_j^2 z_j^{(1,2)}, \tag{3p}$$

$$\sum_{j=1}^{n} \lambda_j^2 z_j^{(2,3)} = \sum_{j=1}^{n} \lambda_j^{31} z_j^{(2,31)} + \sum_{j=1}^{n} \lambda_j^{32} z_j^{(2,32)}, \tag{3q}$$

$$\alpha_i^{32}, \beta_r^{32}, \gamma^{32}, \alpha_i^1, \gamma_y^1, \sigma_c^1, \alpha_i^2, \gamma_x^2, \gamma_y^2, \sigma_c^2, \alpha_i^{31}, \beta_r^{31}, \gamma_y^{31} \geq 0, \ \forall j, i, r, c. \tag{3r}$$

Some descriptions about model (3) are as follows:

1. All the input–output variables in this paper are non-negative so that we choose the direction vector in model (3) as

$$(g_{ik}^{xp}, g_{ck}^{bp}, g_{rk}^{yp}, g_k^{(p,q)}) = (x_{ik}^p, b_{ck}^p, y_{rk}^p, z_k^{(p,q)}), \forall p, q, i, r, c, \tag{4}$$

by according to [25].

2. As we have described in Section 2.1, the indicator of the electricity from other sources (X_2^2) might be equal to zero for some provinces. To prevent the unboundedness issue caused by the zero direction variable, we refer to the method of handling zero data used by Tone and Tsutsui [13] and delete the corresponding input–output inefficiency index from the objective function of

[2] As Fig. 1 shows, there is at most one intermediate product between any two processes in the electric power system. For simplicity, the right subscript of z contains only j.

model (3). Therefore, in the objective function (3a), the item related to X_2^2 is $\sum_{i\in\{i|g_{ik}^{x2}\neq 0\}} \mu_i^{\alpha 2}\alpha_i^2$, not $\sum_{i=1}^{I_2} \mu_i^{\alpha 2}\alpha_i^2$. Other indicators of all the provinces are positive. So, the unboundedness issue does not occur for other indicators.

3. Inequalities (3b)–(3d) are the constraints of inputs, outputs and undesirable outputs with respect to the power generation process; inequalities (3e)–(3h) are those with respect to the transmission and distribution process; inequalities (3i)–(3k) are the input–output constraints with respect to the production power consumption process; inequalities (3l)–(3n) are the input–output constraints with respect to the domestic power consumption process.
4. Every $z_j^{(p,q)}$ is an output of process p and meanwhile it is an input of process q. It links two processes and its value may be increased or decreased from the observed one. Eqs. (3p) and (3q) link two sequential processes, which guarantee the continuity of the intermediate product between the power generation process and the transmission and distribution process, as well as the continuity of the intermediate product between the transmission and distribution process and the power consumption process, respectively. Clearly, both the intermediate products are treated as free links [26] and they may increase or decrease.
5. Equality in (3o) is the constraint of returns to scale (VRS) [27] or damages to scale (DTS) for each process. See [5,28,29] for detailed descriptions on VRS and DTS.

The following theorem proves the feasibility of model (3).

Theorem 1. *Model (3) is feasible.*

Proof. Let $\lambda_k^{'p} = 1, \lambda_j^{'p} = 0, j \neq k, j = 1,\dots,n, \alpha_i^{'p} = \beta_r^{'p} = \gamma^{'p} = \gamma_y^{'p} = \sigma_c^{'p} = \gamma_x^{'p} = 0, \forall p,i,r,c$. Note that we have mentioned that $z_j^{(2,3)} = z_j^{(2,31)} + z_j^{(2,32)}, \forall j$, in Section 2.1. Then, it is easy to see that $(\lambda_j^{'p}, \alpha_i^{'p}, \beta_r^{'p}, \gamma^{'p}, \gamma_y^{'p}, \sigma_c^{'p}, \gamma_x^{'p})$ is a feasible solution of model (3). Therefore, model (3) is feasible. □

Let $\alpha_i^{p*}, \beta_r^{p*}, \sigma_c^{p*}, \gamma_y^{p*}, \gamma_y^{p*}$ be the optimal solution variables of model (3). They are inefficiency measures of the corresponding variables and can be used to judge whether the corresponding input, output and undesirable output are efficient or not. For example, if $\alpha_1^{1*} > 0$, then there exist input excesses in X_1^1 of the DMU k; otherwise, there are no input excesses in X_1^1 of the DMU k. The larger the values of $\alpha_i^{p*}, \beta_r^{p*}, \sigma_c^{p*}, \gamma_x^{p*}, \gamma_y^{p*}$, the worse the efficiency. D_k^* is an inefficiency index and can well reflect all the excesses of inputs and undesirable outputs, as well as output shortages. We judge the overall efficiency of the DMU k according to the following principle: if $D_k^* = 0$, then the DMU k is overall efficient; otherwise, it is overall inefficient; the larger the value of D_k^*, the more inefficient the DMU k. For inefficiency measures, we have the following theorem:

Theorem 2. *For any optimal solution of model (3) with the direction in (4), we have* $0 \le \alpha_i^{p*}, \sigma_c^{p*}, \gamma_x^{p*} \le 1, \ \forall i \in \{i|x_{ik}^p \neq 0\}, \forall c$.

Proof. From the non-negative constraints of model (3), it is clear $\alpha_i^{p*}, \sigma_c^{p*}, \gamma_x^{p*} \ge 0, \ \forall i,c$, establishes. From (4) and the constraint set of model (3), we have $x_{ik}^p - \alpha_i^p g_{ik}^{px} = x_{ik}^p - \alpha_i^p x_{ik}^p \ge \sum_{j=1}^n \lambda_j^p x_{ij}^p \ge 0$ in model (3). Therefore, we have $\alpha_i^* \le 1, \ \forall i \in \{i|x_{ik}^p \neq 0\}$. Similarly, $\sigma_c^{p*}, \gamma_x^{p*} \le 1, \ \forall c$ holds. □

The following indexes is used to measure the inefficiencies of processes.

$$D_k^1 = \sum_{i=1}^{I^1} \mu_i^{\alpha 1}\alpha_i^{1*} + \mu^{\gamma_y 1}\gamma_y^{1*} + \sum_{c=1}^{C_1} \mu_c^{\sigma 1}\sigma_c^{1*}, \tag{5}$$

$$D_k^2 = \sum_{i\in\{i|g_{ik}^{x2}\neq 0\}} \mu_i^{\alpha 2}\alpha_i^{2*} + \mu^{\gamma_x 2}\gamma_x^{2*} + \mu^{\gamma_y 2}\gamma_y^{2*} + \sum_{c=1}^{C^2} \mu_c^{\sigma 2}\sigma_c^{2*}, \tag{6}$$

$$D_k^{31} = \sum_{i=1}^{I^{31}} \mu_i^{\alpha 31}\alpha_i^{31*} + \mu^{\gamma_y 31}\gamma_y^{31*} + \sum_{r=1}^{R^{31}} \mu_r^{\beta 31}\beta_r^{31*}, \tag{7}$$

$$D_k^{32} = \sum_{i=1}^{I^{32}} \mu_i^{\alpha 32}\alpha_i^{32*} + \mu^{\gamma_x 32}\gamma_x^{32*} + \sum_{r=1}^{R^{32}} \mu_r^{\beta 32}\beta_r^{32*}. \tag{8}$$

Obviously, we have

$$D_k^* = \omega^1 D_k^1 + \omega^2 D_k^2 + \omega^3\rho^{31} D_k^{31} + \omega^3\rho^{32} D_k^{32}. \tag{9}$$

Since $\omega^1 + \omega^2 + \omega^3\rho^{31} + \omega^3\rho^{32} = \omega^1 + \omega^2 + \omega^3 = 1$, the system inefficiency is the weighted average of process inefficiencies. Clearly, $D_k^* = 0$ if and only if $D_k^1 = D_k^2 = D_k^{31} = D_k^{32} = 0$. If $D_k^* > 0$, then we can find the specific processes making the system overall inefficiency according to the values of D_k^p.

We notice that the above process inefficiency scores are not always uniquely determined since there may be multiple optimal solutions of model (3), although the overall inefficiency D_k^* is uniquely obtained as the linear program optimum. To find the range in which a process inefficiency may vary, we can solve the maximum and the minimum of D_k^p subject to (3b)–(3r) while keeping the overall inefficiency at the optimal value D_k^*. In our data set with the weights shown in (2), we does not encounter multiple optima and all the above D_k^* and D_k^p are between 0 and 1. Therefore, we use

$$E_k^* = 1 - D_k^*, \quad E_k^p = 1 - D_k^p, \tag{10}$$

to measure the efficiency of DMU k. Note that $\beta_r^{p*}, \gamma_y^{p*} > 1$ might establish for other data sets. In that case, D_k^*, D_k^p might be greater than 1. If this occurs, we can ensure the non-negativity of efficiency indexes by using the measures different from those in (10), such as the SBM [13], setting small enough weights for the inefficiency measures greater than unity, or analyzing performance directly by the inefficiency scores D_k^* and D_k^p. Moreover, if multiple optima are encountered in other data sets, we can adopt the method of Chen et al. [30] to determine process inefficiency.

2.3. DPI and the decomposition

Studying productivity changes of the electric power system and its internal processes can provide a better support for setting policies to improve efficiency. To achieve this purpose, we propose DPI relative to model (3) to measure the productivity changes. For each target DMU k, denote the inefficiency index of process p in period ς with reference to all the DMUs in period τ as $D_k^p(\tau,\varsigma)$, and the inefficiency score of system in period ς with reference to all the DMUs in period τ as $D_k(\tau,\varsigma)$, $\forall p,\tau,\varsigma,k$. According to the green LPI [19] and Eq. (10), we define the system DPI of DMU k from period t to period $t+1$ as

$$DPI_k(t,t+1) = \frac{1}{2}[D_k(t,t) - D_k(t+1,t+1) + D_k(t+1,t) - D_k(t,t+1)], \tag{11}$$

where $D_k(t,t)$ and $D_k(t+1,t+1)$ are calculated by solving model (3), and $D_k(\tau,\varsigma)$ $(\tau \neq \varsigma, \tau,\varsigma = t, t+1)$ is calculated by modifying model (3) as follows: the data of DMU j in the constraints are used in the reference period τ; the relevant observed values of DMU k in the constraints are used in the observed period ς; and since model (3) with different τ and ς is mathematically like the super-efficiency DEA model [31], according to the directional super efficiency model [32], inefficiency measures of all input–output variables are set as free variables to ensure the feasibility, i.e., the constraint set (3r) is deleted.

$DPI_k(t,t+1) > (<,=) 0$ means that the productivity of DMU k from period t to period $t+1$ is progressive (regressive, constant). By referring to [19], let the DSB efficiency change (DEC) index and the DSB technology change (DTC) index of the system be

$$DEC_k(t,t+1) = D_k(t,t) - D_k(t+1,t+1), \tag{12}$$

$$DTC_k(t,t+1) = \frac{1}{2}[D_k(t+1,t+1) - D_k(t,t+1) - D_k(t,t) + D_k(t+1,t)]. \tag{13}$$

Then from (11), (12) and (13), we have

$$DPI_k(t,t+1) = DEC_k(t,t+1) + DTC_k(t,t+1), \tag{14}$$

i.e., the system DPI can be expressed as the sum of the system DEC and the system DTC. System efficiency change is the difference between the system efficiency of DMU k in period t and that in period $t+1$. To be specific, $DEC_k(t,t+1) > (<,=)0$ means the system efficiency of DMU k from period t to period $t+1$ is progressive (regressive, constant). The system technology change index reflects the change in the system production frontier between two periods. $DTC_k(t,t+1) > (<,=)0$ means that system technology from period t to period $t+1$ is progressive (regressive, constant).

Denote the DPI, DEC and DTC of DMU k with respect to process p ($p = 1,2,3,31,32$) from period t to period $t+1$ as $DPI_k^p(t,t+1)$, $DEC_k^p(t,t+1)$ and $DTC_k^p(t,t+1)$, respectively, and define them as

$$DPI_k^p(t,t+1) = \frac{1}{2}[D_k^p(t,t) - D_k^p(t+1,t+1) + D_k^p(t+1,t) - D_k^p(t,t+1)], \tag{15}$$

$$DEC_k^p(t,t+1) = D_k^p(t,t) - D_k^p(t+1,t+1), \tag{16}$$

$$DTC_k^p(t,t+1) = \frac{1}{2}[D_k^p(t+1,t+1) - D_k^p(t,t+1) - D_k^p(t,t) + D_k^p(t+1,t)]. \tag{17}$$

Clearly,

$$DPI_k^p(t,t+1) = DEC_k^p(t,t+1) + DTC_k^p(t,t+1), \ \forall p. \tag{18}$$

So, the DPI of each process can be expressed as the sum of the process efficiency change index and the process technology change index. From model (3), we have

$$D_k(\tau,\varsigma) = \omega^1 D_k^1(\tau,\varsigma) + \omega^2 D_k^2(\tau,\varsigma) + \omega^3 \rho^{31} D_k^{31}(\tau,\varsigma) + \omega^3 \rho^{32} D_k^{32}(\tau,\varsigma), \ \ \forall \tau,\varsigma. \tag{19}$$

Then from (11)–(19), we can easily get the following decomposition:

$$\begin{aligned} &DEC_k(t,t+1) \\ &= \omega^1 DEC_k^1(t,t+1) + \omega^2 DEC_k^2(t,t+1) + \omega^3 \rho^{31} DEC_k^{31}(t,t+1) \\ &+ \omega^3 \rho^{32} DEC_k^{32}(t,t+1), \end{aligned} \tag{20}$$

$$\begin{aligned} &DTC_k(t,t+1) \\ &= \omega^1 DTC_k^1(t,t+1) + \omega^2 DTC_k^2(t,t+1) + \omega^3 \rho^{31} DTC_k^{31}(t,t+1) \\ &+ \omega^3 \rho^{32} DTC_k^{32}(t,t+1), \end{aligned} \tag{21}$$

$$\begin{aligned} &DPI_k(t,t+1) \\ &= \omega^1 DPI_k^1(t,t+1) + \omega^2 DPI_k^2(t,t+1) + \omega^3 \rho^{31} DPI_k^{31}(t,t+1) \\ &+ \omega^3 \rho^{32} DPI_k^{32}(t,t+1) \\ &= \omega^1 DEC_k^1(t,t+1) + \omega^2 DEC_k^2(t,t+1) + \omega^3 \rho^{31} DEC_k^{31}(t,t+1) \\ &+ \omega^3 \rho^{32} DEC_k^{32}(t,t+1) \\ &+ \omega^1 DTC_k^1(t,t+1) + \omega^2 DTC_k^2(t,t+1) + \omega^3 \rho^{31} DTC_k^{31}(t,t+1) \\ &+ \omega^3 \rho^{32} DTC_k^{32}(t,t+1). \end{aligned} \tag{22}$$

That is, $DPI_k(t,t+1)$ is the weighted sum of $DEC_k^p(t,t+1)$ and $DTC_k^p(t,t+1)$.

From the above decomposition, we can analyze the reasons that cause the system's productivity changes. Distinguishing itself from MPI and LPI, the DPI enables DMUs to assess efficiency in a specified direction and considers both inefficiencies in slacks and the actual preferences for each index. The result is a robust efficiency discriminant with broader applicability.

3. Empirical analysis

In this section, the proposed network DEA model and indexes are used to analyze the efficiency as well as the relevant change trends of electric power systems and internal processes of China's provinces. We use Matlab R2014a to solve the proposed network DEA models.

3.1. Data sources

We select 30 provinces in China as DMUs to analyze the efficiencies of their electric power systems. The data ranges from 2013 to 2019. Detailed information on inputs, outputs and intermediate products is shown in Table 1.

Note that CO_2 emissions (B_1^1) are calculated by using the relevant emission factors recommended in the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, combined with various types of energy use data from thermal power generation. The labor index (X_2^{31}) is represented by the number of year-end employment.

For the electricity loss rate (B_1^2), Inner Mongolia is divided into east and west, and Hebei province is divided into north and south power grids. For these two provinces, we take the average value as the electricity loss rate of the province.

In 2013 China Electric Power Yearbook, only the total power consumption data ($Z^{(2,3)}$) of each province is released, while the data of production power consumption ($Z^{(2,31)}$) and domestic power consumption ($Z^{(2,32)}$) are not published. Therefore, we use the regional energy balance table in China Energy Statistical Yearbook to obtain the domestic power consumption in 2013, and then subtract the domestic power consumption from the total power consumption to obtain the production power consumption in 2013.

This value of electricity from other sources (X_2^2) is calculated from the difference between the total power transmission and distribution and thermal power generation (X_1^1). We believe that the total power transmission and distribution is the sum of the total power consumption ($Z^{(2,3)}$) and the electricity loss power. Due to data limitations, we cannot directly obtain the data of electricity loss power, but use the following equality to estimate them.

$$\begin{aligned} Electricity\ loss\ rate &= \frac{Electricity\ loss\ power}{Total\ power\ transmission\ and\ distribution} \\ &= \frac{Electricity\ loss\ power}{Total\ power\ consumption + Electricity\ loss\ power}, \end{aligned} \tag{23}$$

where the data of electricity loss rate (B_1^2) and total power consumption ($Z^{(2,3)}$) can be got from China Electric Power (statistical) Yearbook. Then, the data of electricity loss power are estimated by using (23). When the difference between the total power transmission and distribution and thermal power generation is negative, electricity from other sources (X_2^2) takes a value of zero, the inefficiency measure related with this index is also directly assigned a zero value in the proposed network DEA models. This means that the province's thermal power generation is sufficient to supply the province's power consumption.

The data of thermal power source (X_1^1), other production energy (X_1^{31}), and other domestic energy (X_1^{32}) are obtained from the data of the regional energy balance table in China Energy Statistical Yearbook. There are many types of energy in the energy balance table. Taking into account the collection of specific CO_2 emission coefficients, we do not consider the rarely used energy such as lubricating oil and naphtha, but only unify commonly used energy into standard coal.

Due to the huge amount of data from 2013 to 2019, we only give the descriptive statistics of each indicator in Table 2. Clearly, except (X_2^2), which is equal to zero for some provinces, others are all positive.

3.2. Efficiency analysis

Table 3 shows the efficiency scores and rankings of 30 provinces in 2019. According to the standard of my country's economic zone, the 30 provinces studied in this paper are divided into eastern region, central region and western region. For the convenience of observation, the top 10 and bottom 10 of each efficiency ranking are filled with yellow and pink, respectively. To show the advantages of our new model, the last two column of Table 3 presents the results of the BCC and SBM models with undesirable outputs, which also consider all X, B and Y indicators and the VRS assumption. There are 13 variables in total in the BCC and

Table 1
Data sources.

Indicator	Process				Data resource (2013–2019)
	Power generation	Transmission and distribution	Power consumption		
			Production	Domestic	
Thermal power source X_1^1	Input				China Energy Statistical Yearbook
Power capacity X_2^1	Input				China Electric Power (Statistical) Yearbook[a]
CO_2 emissions B_1^1	Undesirable output				2006 IPCC Guidelines for National Greenhouse Gas Inventories
					China Energy Statistical Yearbook
Thermal power generation $Z^{(1,2)}$	Desirable output	Input			China Electric Power (Statistical) Yearbook
Wire length X_1^2		Input			China Electric Power (Statistical) Yearbook
Electricity from other sources X_2^2		Input			China Electric Power (Statistical) Yearbook
Total consumption $Z^{(2,3)}$		Desirable output			China Electric Power (Statistical) Yearbook
Electricity loss rate B_1^2		Undesirable output			China Electric Power (Statistical) Yearbook
Production electricity consumption $Z^{(2,31)}$		Desirable output	Input		China Electric Power (Statistical) Yearbook
					China Energy Statistical Yearbook
Other production energy X_1^{31}			Input		China Energy Statistical Yearbook
Year-end employment X_2^{31}			Input		Statistical yearbooks of provinces
Fixed asset investment X_3^{31}			Input		China Statistical Yearbook
Gross industrial product Y_1^{31}			Desirable output		China Electric Power (Statistical) Yearbook
Domestic electricity consumption $Z^{(2,32)}$		Desirable output		Input	China Energy Statistical Yearbook
					China Energy Statistical Yearbook
Other domestic energy X_1^{32}				Input	China Energy Statistical Yearbook
Per capita disposable income X_2^{32}				Input	China Statistical Yearbook
Consumption per person Y_1^{32}				Desirable output	China Statistical Yearbook

[a] Due to data limitations, relevant power index data are obtained from the 2014–2019 China Electricity Yearbooks and the 2020 China Electricity Statistical Yearbook.

Table 2
Descriptive statistics of each indicator.

Process	Indicator	Unit	Maximum	Minimum	Standard deviation	Average
Power generation	Thermal power source X_1^1	10^4 tons of standard coal	19 252.62	332.52	3943.44	4837.47
	Power capacity X_2^1	10^4 kWh	10 713	235	2495.85	3478.44
	CO_2 emissions B_1^1	10^4 tons	51 092.13	875.01	10 670.47	13 129.62
	Thermal power generation $Z^{(1,2)}$	10^8 kWh	48 240 000	1 040 000	11 919 782.26	15 061 523.81
Transmission and distribution	Wire length X_1^2	kilometer	119 677	8602	27 595.25	57 488.10
	Electricity from other sources X_2^2	10^8 kWh	40 529 360.06	0	7 799 812.02	6 728 762.93
	Electricity loss rate B_1^2	%	12.92	1.42	1.70	6.09
	Total power consumption $Z^{(2,3)}$	10^8 kWh	66 958 511	2 320 000	14 375 935.89	20 499 614.78
Production power consumption	Other production energy X_1^{31}	10^4 tons	98 289.70	1350.77	8636.38	8318.24
	Year-end employment X_2^{31}	10^4	6995	314.21	1688.83	2663.13
	Fixed asset investment X_3^{31}	10^8 RMB	59 073.84	2361.10	13 398.11	19 592.45
	Gross industrial product Y_1^{31}	10^8 RMB	107 671.07	2122.06	20 592.20	26 477.68
	Production power consumption $Z^{(2,31)}$	10^8 kWh	56 171 946	1 954 400	12 658 997.44	17 758 783.14
Domestic power consumption	Other domestic energy X_1^{32}	10^4 tons	2222.95	60.21	455.04	693.80
	Per capita disposable income X_2^{32}	RMB/per person	69 441.60	10 954.40	10 655.27	24 451.26
	Consumption per person Y_1^{32}	RMB/per person	45 605.10	8288	6959.40	17 542.29
	Domestic power consumption $Z^{(2,32)}$	10^8 kWh	10 786 565	193 400	2 010 420.15	2 740 831.63

SBM models. It is not difficult to see that the efficiency identification ability of both the model is very poor, and all provinces except Hebei and Shandong are efficient. However, our model has a strong ability to identify the efficiency, only four provinces are systematically efficient. In addition, our model can also give the efficiency of four internal processes.

As can be seen from Table 3, among the top 10 provinces in system efficiency, Jiangsu, Zhejiang, Hainan and Qinghai is system efficient. Their efficiency score of each process is equal to 1. All of the inefficient provinces ranked top 10 only have one inefficient process. Guangdong's power generation efficiency is not good, resulting in its inefficiency in terms of the entire system. The domestic power consumption process of Beijing, Shandong and Inner Mongolia are inefficient, ranking 17th, 18th and 28th, respectively. Shanghai and Ningxia perform badly in the power generation process and the corresponding process efficiency scores rank 20th and 22th, respectively. These provinces need to improve the system efficiency by improving the efficiency of the corresponding inefficient processes. Among the bottom 10 provinces in the system efficiency, three processes in Heilongjiang, Jilin, Chongqing, Yunnan and Shaanxi rank in the bottom 10, while four processes in Hebei all rank in the bottom 10. These provinces should learn from the experience of efficient provinces to improve the efficiency of their electric power systems.

In order to improve the system efficiency of each province, we can refer to the inefficiency measure of each indicator in the proposed network DEA model. We respectively select the two provinces with the lowest power system efficiency in eastern, central and western regions in 2019, as well as Guangdong and Beijing provinces, both of which have high system efficiency scores in 2019 but are inefficient, as representatives to analyze the inefficiency measures (i.e., α_i^{p*}, β_r^{p*}, σ_c^{p*}, γ_x^{p*}, γ_y^{p*}). Table 4 shows these inefficiency measures. It can be seen from Table 4 that the system inefficiency in Guangdong is caused by the inefficiency of thermal power source, power capacity and CO_2 emissions from the power generation process. Among them, the inefficiency of power capacity is the largest. The inefficiency in Beijing comes from the domestic power consumption process and is caused by the inefficiency of other domestic energy and per capita disposable income, among which the inefficiency of other domestic energy is larger. Therefore, Guangdong and Beijing can focus on power capacity and other domestic energy, respectively, so as to improve the system efficiency. Hebei and Tianjin are the two provinces with the lowest system efficiency in the eastern region. Most of the indicators in Hebei's power generation process, transmission and distribution process and production consumption process are inefficient. Tianjin's transmission and distribution process and domestic power consumption process are efficient, and most indicators in the other two processes need to be improved. Jilin and Heilongjiang are the two provinces with the lowest system efficiency in the central region, and moreover, Heilongjiang is the provinces with the lowest system efficiency in China. The two provinces have the same inefficiency indicators in the processes of

Table 3
Efficiency scores and rankings in 2019.

	Province	Efficiency					Rank					BCC	SBM
		E^*	E^1	E^2	E^{31}	E^{32}	E^*	E^1	E^2	E^{31}	E^{32}		
Eastern	Beijing	0.985	1.000	1.000	1.000	0.913	6	1	1	1	17	1.000	1.000
	Tianjin	0.893	0.818	1.000	0.723	1.000	17	24	1	16	1	1.000	1.000
	Hebei	0.792	0.848	0.806	0.604	0.837	29	21	27	25	23	0.894	0.472
	Liaoning	0.896	0.796	0.991	1.000	0.805	15	25	20	1	27	1.000	1.000
	Shanghai	0.950	0.850	1.000	1.000	1.000	9	20	1	1	1	1.000	1.000
	Jiangsu	1.000	1.000	1.000	1.000	1.000	1	1	1	1	1	1.000	1.000
	Zhejiang	1.000	1.000	1.000	1.000	1.000	1	1	1	1	1	1.000	1.000
	Fujian	0.896	0.931	0.860	0.795	1.000	16	13	22	15	1	1.000	1.000
	Shandong	0.985	1.000	1.000	1.000	0.907	7	1	1	1	18	0.945	0.663
	Guangdong	0.989	0.968	1.000	1.000	1.000	5	12	1	1	1	1.000	1.000
	Hainan	1.000	1.000	1.000	1.000	1.000	1	1	1	1	1	1.000	1.000
Central	Shanxi	0.915	0.904	1.000	0.901	0.783	13	17	1	13	30	1.000	1.000
	Jilin	0.820	0.737	1.000	0.597	0.850	26	29	1	27	22	1.000	1.000
	Heilongjiang	0.761	0.788	0.797	0.532	0.867	30	27	28	30	19	1.000	1.000
	Anhui	0.921	1.000	1.000	0.664	0.860	12	1	1	22	20	1.000	1.000
	Jiangxi	0.909	1.000	1.000	0.644	0.807	14	1	1	23	26	1.000	1.000
	Henan	0.871	0.908	0.845	0.722	1.000	19	16	23	17	1	1.000	1.000
	Hubei	0.869	1.000	0.788	0.700	0.935	20	1	29	18	16	1.000	1.000
	Hunan	0.823	0.856	0.817	0.641	0.949	24	19	24	24	15	1.000	1.000
Western	Inner Mongolia	0.967	1.000	1.000	1.000	0.799	8	1	1	1	28	1.000	1.000
	Guangxi	0.878	0.856	1.000	0.557	1.000	18	18	1	28	1	1.000	1.000
	Chongqing	0.803	0.793	0.862	0.682	0.827	27	26	21	20	24	1.000	1.000
	Sichuan	0.860	0.739	1.000	0.686	1.000	21	28	1	19	1	1.000	1.000
	Guizhou	0.833	0.916	0.814	0.541	1.000	23	15	26	29	1	1.000	1.000
	Yunnan	0.802	0.681	1.000	0.599	0.854	28	30	1	26	21	1.000	1.000
	Shaanxi	0.822	1.000	0.728	0.675	0.799	25	1	30	21	29	1.000	1.000
	Gansu	0.856	0.822	0.814	0.867	1.000	22	23	25	14	1	1.000	1.000
	Qinghai	1.000	1.000	1.000	1.000	1.000	1	1	1	1	1	1.000	1.000
	Ningxia	0.947	0.840	1.000	1.000	1.000	10	22	1	1	1	1.000	1.000
	Xinjiang	0.944	0.929	1.000	1.000	0.809	11	14	1	1	25	1.000	1.000
Eastern		0.944	0.928	0.969	0.920	0.951							
Central		0.861	0.899	0.906	0.675	0.881							
Western		0.883	0.870	0.929	0.782	0.917							

power generation, production power consumption and domestic power consumption. Therefore, in the processes of power generation and consumption, the two provinces can design similar policies to improve their efficiency. In the transmission and distribution process, Jilin is efficient, while except the indicator of total electricity distribution, others of Heilongjiang in this process are inefficient. Yunnan and Chongqing are the two provinces with the lowest system efficiency in the western region. Yunan performs well in the transmission and distribution process, no indicator is inefficient in this process, but most indicators in other processes need to be improved greatly. More than half of indicators of Chongqing in each process have positive inefficiency measures, so Chongqing can be greatly improved in many aspects.

In order to analyze the spatial distribution of system efficiency, we use Python software to draw the efficiency distribution map of the power system and its processes in 2019. The detailed results are shown in Figs. 2 and 3. It can be seen from Fig. 2 that the power system efficiency in the eastern coastal region is relatively high, and those in the central and western regions are relatively low. There are several provinces in the western region with relatively high system efficiency scores, but most provinces are less efficient. Provinces colored in blue are the most inefficient ones in the map, which shows that there is more room for improving in the system efficiency of these provinces. The government should formulate policies to improve their efficiencies, such as eliminating outdated equipment, promoting energy conservation, reducing emission, and so on.

Fig. 3 is the spatial distribution map of the process efficiency in 2019. For the power generation process, it can be seen from Fig. 3(a) that the provinces with higher power generation efficiency are mainly in the eastern China and the central provinces close to the east. This might be because the eastern region is economically developed and has advantages in technology and talents, while the central provinces close to the east can more easily learn advanced technologies. Provinces with poor power generation performance are mainly in southwestern China and northeast China, which are shown in blue. Inefficient provinces in these two regions should learn from the experience and technology of eastern China to improve their power generation efficiency, and at the same time develop wind energy, solar energy and other power generation technologies according to their own natural environment advantages.

For the power transmission and distribution process, it can be seen from Fig. 3(b) that the advantages of the eastern region are still obvious. Except Hebei, Liaoning and Fujian, other eastern provinces have relatively high power transmission and distribution efficiency. The efficiency of power transmission and distribution in the western region is polarized. Qinghai, Sichuan and Yunnan have good performance, but central and western provinces close to the central region, such as Shaanxi and Hubei, etc., have poor performance. It may be because the rugged terrain in these regions is relatively difficult to build power grids. Therefore, in order to improve the transmission and distribution efficiency, it is necessary to carry out scientific planning for the construction of the power grid by considering their actual situations.

For the power consumption process, it can be seen from Figs. 3(c) and 3(d) that the two spatial distribution maps have great differences, especially the efficiency of the two power consumption processes in the central and western regions, which indicates that it is necessary

Table 4
Inefficiency measures of typical provinces.

Process	Indicator	Eastern		Central		Western		Eastern	
		Hebei	Tianjin	Jilin	Heilongjiang	Yunnan	Chongqing	Guangdong	Beijing
Power generation	Thermal power source	0.221	0.313	0.488	0.402	0.501	0.328	0.066	0.000
	Power capacity	0.011	0.001	0.036	0.011	0.429	0.090	0.109	0.000
	CO_2 emissions	0.383	0.414	0.556	0.443	0.653	0.476	0.029	0.000
	Thermal power generation	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Transmission and distribution	Wire length	0.464	0.000	0.000	0.530	0.000	0.587	0.000	0.000
	Electricity from other sources	0.035	0.000	0.000	0.130	0.000	0.020	0.000	0.000
	Electricity loss rate	0.471	0.000	0.000	0.357	0.000	0.084	0.000	0.000
	Total power consumption	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Production power consumption	Other production energy	0.645	0.361	0.358	0.462	0.357	0.269	0.000	0.000
	Year-end employment	0.000	0.000	0.233	0.251	0.265	0.217	0.000	0.000
	Fixed asset investment	0.274	0.491	0.479	0.387	0.429	0.591	0.000	0.000
	Gross industrial product	1.058	0.532	0.944	1.241	0.955	0.513	0.000	0.000
	Production activity electricity consumption	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Domestic power consumption	Other domestic energy	0.641	0.000	0.599	0.534	0.540	0.691	0.000	0.338
	Per capita disposable income	0.000	0.000	0.000	0.000	0.000	0.007	0.000	0.038
	Consumption per person	0.010	0.000	0.000	0.000	0.043	0.000	0.000	0.000
	Domestic activity electricity consumption	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

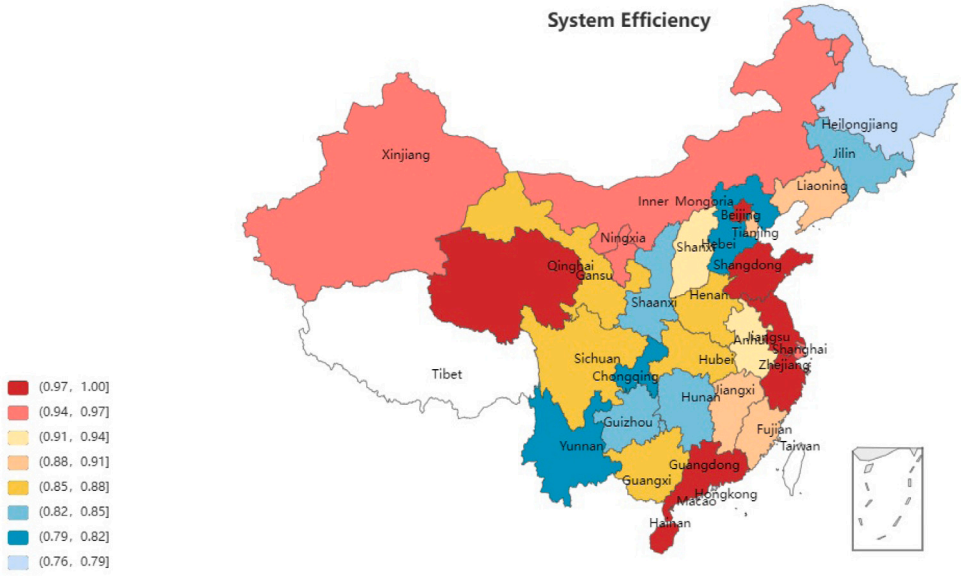


Fig. 2. Efficiency spatial distribution map of the entire power system.

to evaluate the domestic consumption process and the production consumption process separately. It can be seen from Figs. 3(c) and 3(d) that between the two power consumption processes, the domestic consumption efficiency in eastern China is relatively high, while the production power consumption efficiency of some eastern provinces, such as Fujian and Tianjin, is not good. The production consumption process in the northwest region performs well but the domestic consumption process in this region performs poorly. Most provinces in the central and southwestern regions have lower production consumption efficiency but higher domestic consumption efficiency. The northwest region occupies a large area and is rich in natural resources, so it has certain advantages in the production consumption process. However, the northwest region is relatively remote and economically backward, and people's living standards are not high, so the efficiency of the domestic consumption process is low. In recent years, a lot of energy-consuming industries in eastern China have been transferred to the central and southwestern regions, which make the local economy and people's living standards improved, so the domestic consumption process in the central and southwestern regions is more efficient.

3.3. Productivity analysis

Table 5 shows the average values of DPI, DEC and DTC in each region, where the values greater (or less) than zero are filled with green (or pink). After the system productivity of each region suffers from a large decline from 2014 to 2015, both the eastern and central regions show relatively small gains in system productivity. For the western region, after experiencing an increasingly large productivity decline from 2013 to 2016 and from 2017 to 2018, it shows significant system productivity improvement from 2016 to 2017 and from 2018 to 2019 due to the technology progress. The western region can summarize the successful experience from 2016 to 2017 and from 2018 to 2019, to help maintain the progress of system productivity and improve the system efficiency.

Since Table 5 shows the average DPI values of three regions all decline from 2014 to 2015, we draw scatter Fig. (shown in Fig. 4) for DEC and DTC of processes in provinces from 2014 to 2015. As can be seen from Fig. 4, in the power generation process as well as the transmission and distribution progress, most provinces show

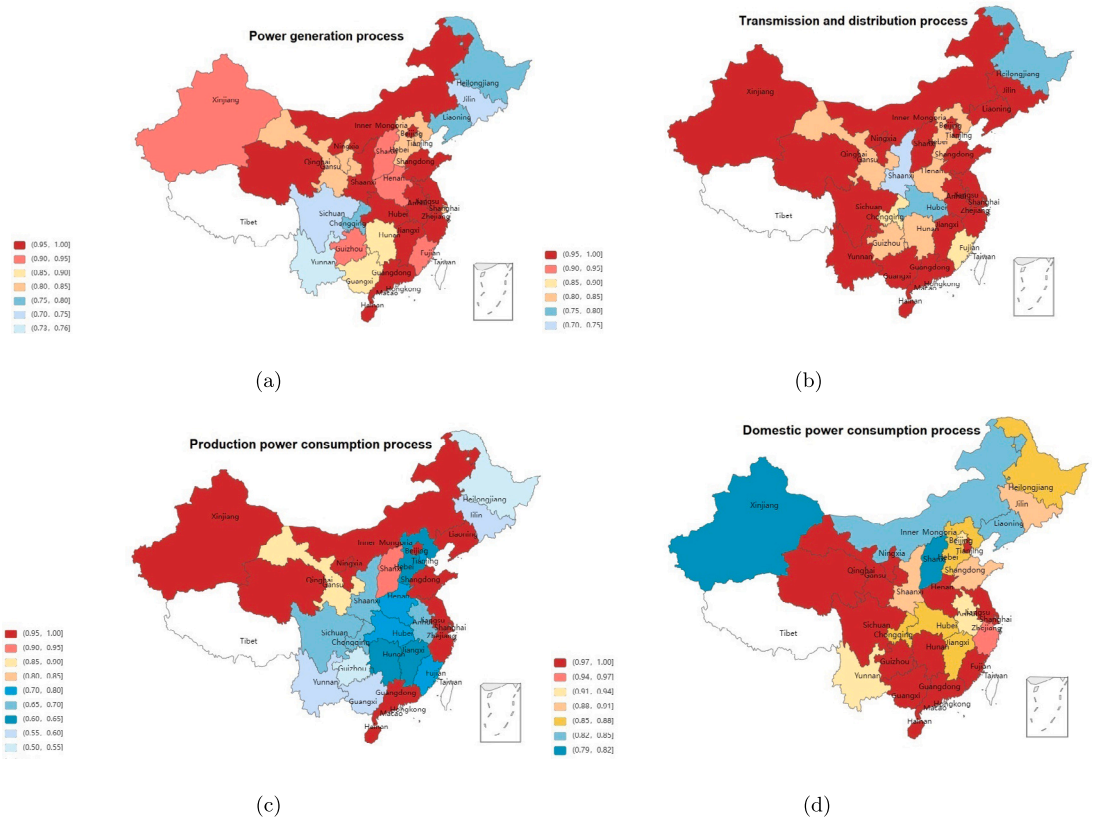


Fig. 3. Efficiency spatial distribution map of different processes.

Table 5
DPI, DEC and DTC of the system.

Region	2013–2014			2014–2015			2015–2016			2016–2017			2017–2018			2018–2019		
	DPI	DEC	DTC	DPI	DEC	DTC	DPI	DEC	DTC	DPI	DEC	DTC	DPI	DEC	DTC	DPI	DEC	DTC
Eastern	0.021	0.026	−0.005	−0.025	−0.019	−0.006	0.010	0.047	−0.037	0.013	−0.014	0.027	0.019	−5.8E−05	0.019	0.016	−0.002	0.018
Central	0.015	0.015	−1.9E−04	−0.019	−0.021	0.002	0.004	0.061	−0.057	0.014	−0.029	0.043	0.027	0.039	−0.012	0.005	−0.008	0.014
Western	−0.007	0.017	−0.024	−0.013	−0.010	−0.003	−0.077	0.065	−0.142	0.075	−0.039	0.114	−0.059	0.024	−0.083	0.007	−0.014	0.021

technology progress with efficiency regress. In the production power consumption progress, all the provinces in the central region show both technology and efficiency regress, while the technology of many provinces in the eastern and western regions are regressive but their efficiency is progressive. In the domestic power consumption process, most provinces in the central region show technology progress but their efficiency is decline, while in the eastern and western regions, most provinces show technology progress, but some of them show efficiency progress and some show efficiency regress. In summary, the productivity regress of system in most provinces from 2014 to 2015 is caused by the technology regress in the production power consumption process and the efficiency regress in the power generation process as well as the transmission and distribution progress.

In order to better understand the change of the system productivity, we analyze DEC and DTC of each process. Figs. 5 and 6 are the line graphs of the DEC and DTC changes of the four processes in each region, respectively.

As can be seen from Fig. 5, the efficiency in the eastern, central and western regions has great progress in the power generation process from 2013 to 2014, but there is no significant change in the following years. In the transmission and distribution process, the efficiency change trend of the eastern, central and western regions is consistent. The average DEC gradually increases from 2013 to 2016, abruptly worsens in 2017, improves again in 2018, and finally shows a very slight change in 2019. In the production power consumption process, the efficiency changes in the eastern and central regions are relatively small and show a decline in most years, while the efficiency in the western region changes dramatically. The average DEC of western provinces gradually improves from 2013 to 2016, then increases again from 2017 to 2018 after a sharp deterioration from 2016 to 2017, but finally shows a decline from 2018 to 2019. In the domestic power consumption process, the efficiency change is small and does not exceed 0.02 before 2016. From 2013 to 2017, the efficiency of the eastern region gets better and better, but the efficiency drops sharply in 2018 and continues to show a slight downward trend in 2019. Except for a slight decrease in efficiency from 2013 to 2015, the efficiency changes in the central and western regions are consistent with that in the eastern region from 2016 to 2018, and the efficiency of the two regions maintains a slight increase in 2019.

We can see from Fig. 6 that in the power generation process, all regions in China experience technology regression from 2013 to 2014. Since 2014, there has been no significant technology change in the eastern and central regions, while the technology change in the western region fluctuate greatly. From 2014 to 2019, technology in the western region experiences slight progress, regress, progress, regress and slight progress in turn. In the transmission and distribution process, the technology change trends in the eastern, central and western regions are nearly consistent, and show progress to regress, to progress, to regress (except the eastern region) and then to slight progress. In the production power consumption process, the technology change

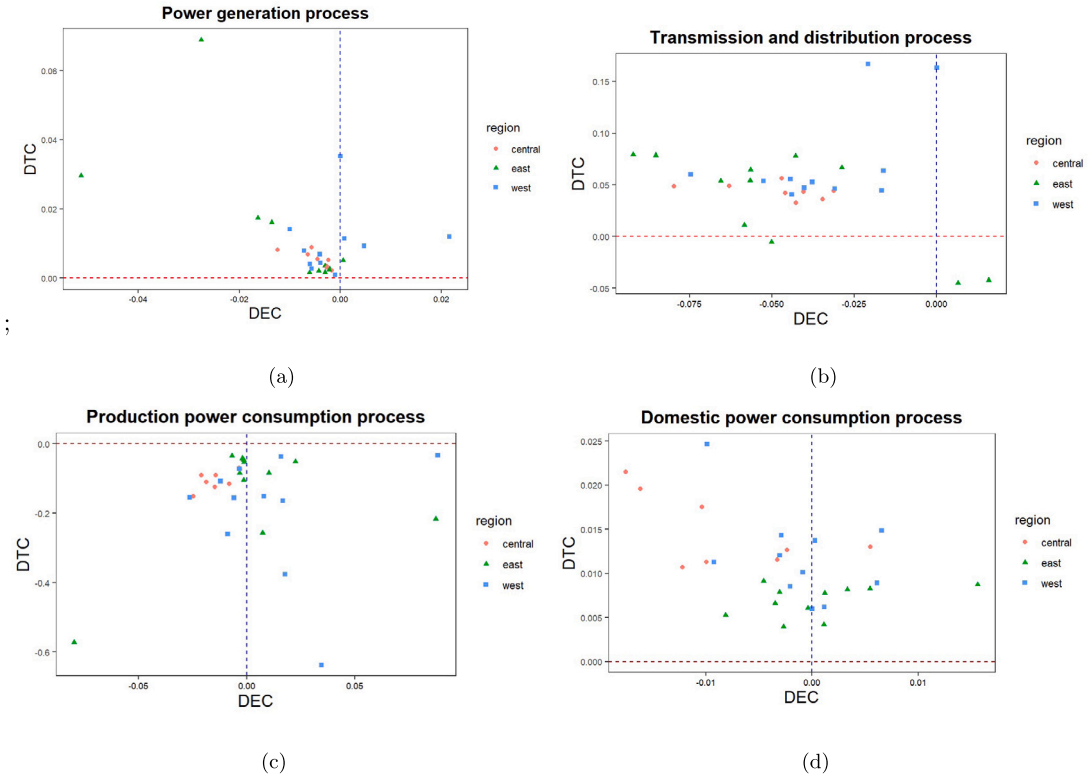


Fig. 4. DEC and DTC of provincial processes in 2014–2015.

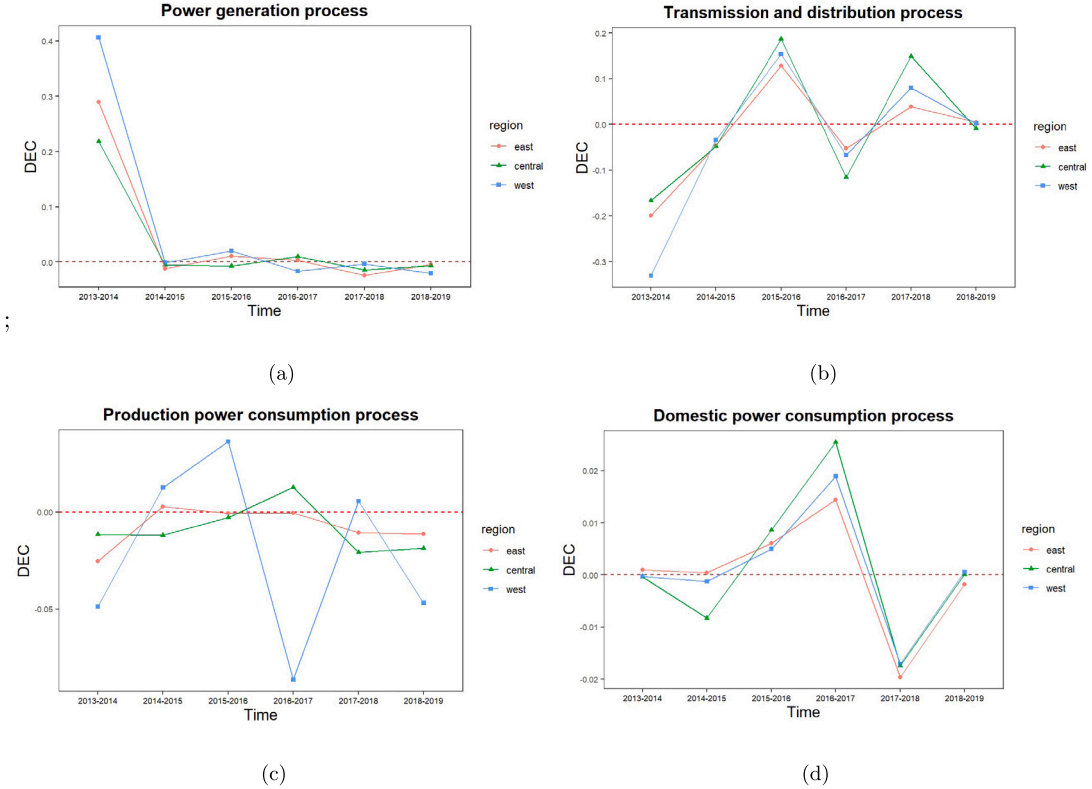


Fig. 5. Average DEC of processes in each region.

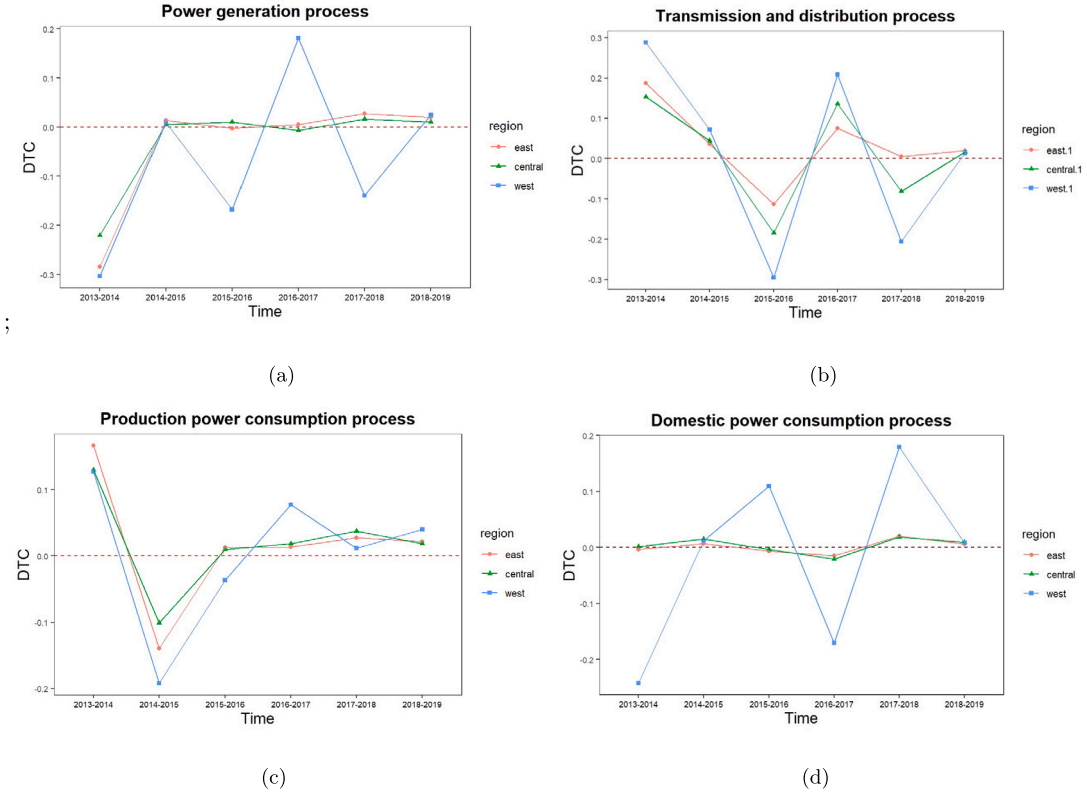


Fig. 6. Average DTC of processes in each region.

trends of eastern and central regions are also almost consistent. The technology change trend of the western region is V-shaped, and the technology regress is the largest in 2014–2015. In the domestic power consumption process, the technology changes in the eastern and central regions remain nearly unchanged, while the technology change in the western region is M-shaped, with the greatest technology progress from 2017 to 2018.

Combining Figs. 5 and 6, it can be seen that both efficiency and technology changes of each process in eastern China are nearly the same as those in the central China. From 2018 to 2019, the efficiency of the production power consumption process in the three regions is obviously regressive, but the corresponding technology change is progressive. The efficiency and technology of other processes in the three regions do not change significantly from 2018 to 2019. Therefore, it is necessary to formulate the corresponding production power consumption policy to improve productivity.

4. Conclusions

In this paper, we innovatively decompose the complex electric power system into the serial processes of power generation, transmission and distribution, and power consumption, where the power consumption process is divided into two parallel sub-processes of production power consumption and domestic power consumption. Then, we establish a network DSB model to evaluate the efficiency of electric power systems and internal processes. To evaluate productivity changes, we propose the DPI. The system DPI can be decomposed into the weighted sum of DEC and DTC indexes of all the progresses, and then the reasons for productivity changes of electric power systems can be analyzed in more detail. The proposed network DEA model and indexes are applied to evaluate the electric power system efficiency of 30 provinces in China from 2013 to 2019, and the regional characteristics and changes of system efficiency are analyzed.

The empirical results show that the efficiency of China's electric power system is high in the eastern coastal region. Many eastern provinces have the good efficiency in all the processes. Most central provinces perform better than western ones in the processes of power generation and domestic power consumption, but many western provinces perform well in the processes of transmission and distribution as well as production power consumption. Central and eastern regions nearly show the same trend of system productivity. Except from 2014 to 2015, the system productivity in central and eastern regions is improved. The system productivity of the western region is backward from 2013 to 2016 and from 2017 to 2018, but makes great progress from 2016 to 2017 and from 2018 to 2019. The efficiency change trends of each process in the eastern region is also nearly as same as that in the central region. From 2018 to 2019, the efficiency of the production power consumption process in eastern, central and western regions is regressive, but the technology is progressive. The efficiency and technology of other processes do not change significantly in three regions. Local governments should formulate corresponding policies according to these results to improve the productivity of electric power systems.

In future research work, we can carry out the following aspects:

First, it can be seen that the efficiency of systems and processes of 30 provinces cannot be fully ranked. Therefore, when establishing the network DEA model, the super-efficiency DEA model and the cross-efficiency model can be considered to improve the discrimination of efficiency.

Secondly, we do not consider the effect of carry-over variables when studying the efficiency changes of electric power system between two periods of time. We can consider using dynamic network DEA models which can consider carry-over variables to evaluate the efficiency of electric power system and analyze the corresponding productivity changes.

Finally, as we pointed out in Section 2.2, one of the disadvantages of our approach is that it is possible to generate inefficiency measures

greater than 1 for outputs of a process. In future research, we can make all the inefficiency measures less than 1 by selecting other direction vectors, such as those in the RAM model [33].

CRediT authorship contribution statement

Ruiyue Lin: Conceptualization, Formal analysis, Investigation, Methodology, Project administration, Validation, Writing – original draft. **Lingling Ding:** Conceptualization, Data curation, Methodology, Software. **Zongxin Li:** Conceptualization, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

The work was supported by the National Natural Science Foundation of China [grant number 71971163], Ministry of Education of the People's Republic of China Humanities and Social Sciences Youth Foundation [grant number 23YJCZH116], the Annual Project of Shaanxi Provincial Social Science Foundation, China [grant number 2023D014], the Scientific Research Project of Shaanxi Provincial Department of Education, China [grant number 23JK0220], Graduate Scientific Research Foundation of Wenzhou University [grant number 3162023003035], and Northwest University National Social Science Incubation Project [grant number 22XNFH012]. We would like to express our gratitude to two anonymous referees for their constructive comments, which have helped us to significantly improve the paper.

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